

# LISA without Frontiers: Global Fits, Signal Catalogues, Astrophysical Backgrounds

Machine Learning for Galactic Binary Search & Parameter Estimation

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# Sprint Goals

1. Identify **key challenges** for ML in GB search/PE
2. Define **concrete projects** we can start in 2.5 days
3. Establish **long-term directions** for follow-up collaboration
4. Build **shared infrastructure**: datasets, benchmarks, interfaces

## Philosophy

Tool-agnostic: we focus on the *science challenges*, not any single method.

ML methods are means to an end – the end is LISA-ready GB analysis.

# The Galactic Binary Challenge

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# What Makes GBs Hard?

## Signal characteristics:

- Quasi-monochromatic:  $f$  evolves slowly ( $\dot{f} \sim 10^{-15}$ – $10^{-8}$  Hz/s)
- 7–8 parameters per source:  $(f, \dot{f}, \mathcal{A}, \iota, \psi, \varphi_0, \lambda, \beta)$
- Long-lived: most persist for entire mission duration
- Overlapping:  $\mathcal{O}(10^4)$  resolvable,  $\mathcal{O}(10^7)$  total in band

## The confusion problem:

- Below  $\sim 3$  mHz: unresolved GBs form confusion foreground
- Sources separated by  $\Delta f < 1/T_{\text{obs}}$  blend together
- “Resolvable” depends on what else is in the band

# Parameter Degeneracies

## Intrinsic degeneracies:

- Amplitude  $\leftrightarrow$  inclination  
 $\mathcal{A} \propto \mathcal{M}_c^{5/3} f^{2/3} / D_L \times g(\iota)$
- Distance completely degenerate without EM counterpart

## Extrinsic degeneracies:

- Sky location: rings/arcs from Doppler modulation
- Polarization angle  $\psi$  correlates with phase

## Inter-source correlations:

- When  $\Delta f < 1/T$ , likelihood couples nearby sources
- Correlated posteriors between neighbors
- Error in one source biases others

## Current Approaches

| Method                | Strengths           | Limitations                |
|-----------------------|---------------------|----------------------------|
| Iterative subtraction | Simple, fast        | Biased, error accumulation |
| F-statistic search    | Fast detection      | PE requires follow-up      |
| RJMCMC (GLASS, Eryn)  | Rigorous, trans-dim | Slow, scales poorly        |
| Nested sampling       | Evidence estimation | Fixed dimension            |

### The Gap (?)

While fast methods are often desirable, it is not yet clear whether they are strictly necessary for the global fit. Understanding this trade-off, and whether machine-learning approaches can contribute meaningfully, remains an open question.

## Long-Term Challenges

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# Challenge 1: The Global Fit

**The defining problem:** Jointly infer  $\mathcal{O}(10^4)$  sources + their number + confusion foreground

## Key difficulties:

- Trans-dimensional: unknown number of sources
- Combinatorial: which peaks are real sources?
- Correlated: nearby sources share likelihood support
- Computational:  $\mathcal{O}(K)$  parameters, but  $\mathcal{O}(K^2)$  correlations?

## ML opportunities:

- Amortized trans-dimensional inference
- Learned proposals for efficient RJMCMC jumps
- Surrogate likelihoods for faster evaluation
- Hierarchical: coarse global  $\rightarrow$  refined local



## Challenge 2: Source Confusion & Correlation

When  $\Delta f < 1/T$ , sources are **not separable** in likelihood.

### Questions:

- Can we learn to identify “confused” vs “clean” sources?
- How to represent uncertainty over source assignments?
- Joint posteriors vs marginalized single-source posteriors?
- When does factorized inference fail catastrophically?

### Fundamental Limit?

Is there an information-theoretic limit to separability, or can clever methods do better than naive  $1/T$  resolution?

## Challenge 3: Verification Binaries & EM Synergy

~50 known GBs with EM counterparts (AM CVn, detached WDs)

Why they matter:

- **Guaranteed detections** with known sky positions
- **Validation anchors:** do our methods recover known parameters?
- **Science targets:** GW + EM  $\rightarrow$  distance ladder, binary physics

For ML:

- Natural test set with ground truth
- Can we use EM priors to improve PE?
- Semi-supervised: known sources + unknown sources together

## Challenge 4: Confusion Foreground

Below the resolution limit, unresolved GBs act as **colored noise**.

### Modeling approaches:

- Analytic: Galactic population models  $\rightarrow$  predicted PSD
- Empirical: fit foreground spectrum from data
- Iterative: resolve sources  $\rightarrow$  update foreground  $\rightarrow$  resolve more

### ML considerations:

- How to train with foreground? Fixed PSD vs learned?
- Can we learn the foreground jointly with source catalog?
- Domain shift: training foreground  $\neq$  real foreground

## Challenge 5: From Detection to Catalog

The science output is a **catalog with uncertainties**.

### Catalog requirements:

- Per-source posteriors (or summary statistics)
- Completeness and purity estimates
- Flags for confused/uncertain sources
- Format compatible with population inference

### Open questions:

- What SNR threshold defines “detection”?
- How to handle sources we’re not sure are real?
- Selection function: what did we miss?

## Sprint Projects

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# Project Tiers

**Tier 1:** Concrete & achievable in 2.5 days

**Tier 2:** Foundational work – start now, continue after

**Tier 3:** Exploratory – scope, design, roadmap

## Team formation:

- Mix of ML expertise + GB/LISA expertise per team
- 3–5 people per project
- Check-ins at end of each day

## Tier 1: Concrete & Achievable

| Project                              | Description                                                                                           |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| Verification binary recovery         | Test ML methods on $\sim 50$ known GBs – do we recover correct parameters?                            |
| Confusion regime mapping             | Characterize where (in $f$ , SNR, $\Delta f$ space) methods fail                                      |
| GB-realistic data generator          | Lightweight tool for $(f, \dot{f}, \mathcal{A}, \iota, \psi, \varphi_0, \lambda, \beta) + \text{TDI}$ |
| ML $\rightarrow$ MCMC initialization | Use fast inference to warm-start GLASS/Eryn                                                           |
| Catalog-level metrics                | Define: what makes a good GB catalog?                                                                 |

## Tier 2: Foundational Work

| Project                          | Description                                              |
|----------------------------------|----------------------------------------------------------|
| Frequency separation study       | At what $\Delta f$ do methods break? Theory vs empirical |
| Inclination–amplitude degeneracy | Can ML break this better than uninformative priors?      |
| Confusion foreground modeling    | How to train with/around unresolved background           |
| Sky localization accuracy        | ML vs F-statistic vs full PE – where does each win?      |



## Tier 3: Towards the Global Fit

| Project                          | Description                                                         |
|----------------------------------|---------------------------------------------------------------------|
| Hierarchical architecture design | Sketch how to scale to $\mathcal{O}(10^3\text{--}10^4)$ sources     |
| Correlation-aware posteriors     | Move beyond factorized per-source inference                         |
| Iterative ML refinement          | Detect $\rightarrow$ subtract $\rightarrow$ detect pipeline with ML |
| LDC integration roadmap          | Path to Sangria/Radler participation                                |

## Data & Benchmarks

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# Dataset Complexity Ladder

| Level | Signal Model                      | Noise          | Sources       | Status |
|-------|-----------------------------------|----------------|---------------|--------|
| 0     | Monochromatic sinusoids           | White Gaussian | $K \leq 10$   | ✓      |
| 1     | + frequency derivative $\dot{f}$  | White Gaussian | $K \leq 10$   | —      |
| 2     | + sky location $(\lambda, \beta)$ | White Gaussian | $K \leq 10$   | —      |
| 3     | Full 8-param GBs                  | Stationary PSD | $K \leq 50$   | —      |
| 4     | Full GBs + TDI response           | Stationary PSD | $K \leq 100$  | —      |
| 5     | + confusion foreground            | Realistic      | $K \sim 10^3$ | —      |
| 6     | LDC Sangria/Radler                | Full realism   | Full catalog  | —      |

## Workshop Goal

Agree on Level 1–3 specifications, build shared data generation tools.

# Benchmark Design Principles

## What to measure:

- **Cardinality accuracy:** correct number of sources?
- **Parameter recovery:** bias, variance, Wasserstein distance
- **Calibration:** are posteriors honest?
- **Computational cost:** time, memory, scaling with  $K$

## Test case design:

- Vary: SNR,  $K$ ,  $\Delta f_{\min}$ , noise level
- Include: easy (isolated, loud) + hard (confused, faint)
- Adversarial: edge cases, out-of-distribution

**Deliverable:** Shared benchmark suite the community can use.

## Discussion

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## Key Questions for Participants

1. What do you need from a GB catalog for **your science**?
2. Where does your current pipeline **bottleneck**?
3. What would “good enough” ML look like for GBs?
4. Verification binaries: validation set or science target?
5. How do we handle sources we’re **not sure are real**?
6. Should we aim for **one community tool** or a **diverse ecosystem**?

# Workshop Schedule (Draft)

## Day 1 (Half day intro + project kickoff)

- Morning: GB challenge overview, existing methods landscape
- Afternoon: Form teams, scope projects, set up environments

## Day 2 (Hands-on hacking)

- Full day of project work with check-ins
- Evening: Progress share, cross-pollination

## Day 3 (Morning wrap-up)

- Final presentations (15 min each)
- Discussion: lessons learned, path forward
- Define follow-up collaborations

## Form teams & choose projects

### Resources:

- Shared code repository: [TBD]
- Data generation tools: [TBD]
- Slack/Discord channel: [TBD]

Questions? Ideas? Concerns?